

HAT-P-18b

R-2259

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-18_140303 · created 2026-08-16T01:11:09+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456719.94641 +/- 0.0009 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.115 +/- 0.053
Transit depth (Rp/Rs)^2	1.32 +/- 1.22 [%]
Orbital Inclination (inc)	86.87 +/- 1.86
Airmass coefficient 1 (a1)	1.049 +/- 0.036
Airmass coefficient 2 (a2)	-0.027 +/- 0.012
Scatter in the residuals of the lightcurve fit is	3.49 %
Best Comparison Star	#3 - [275, 78]
Optimal Aperture	2.77
Optimal Annulus	11.07
Transit Duration (day)	0.0 +/- 0.0

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.115 $\pm$ 0.053 vs 0.1356 (deviation 0.39 σ)
Duration fit vs published	0.0 d vs 0.1129 d (deviation inf σ)
Mid-time O-C	1.1 min
Depth SNR	2.2
Residual scatter	3.49 %

Light curve

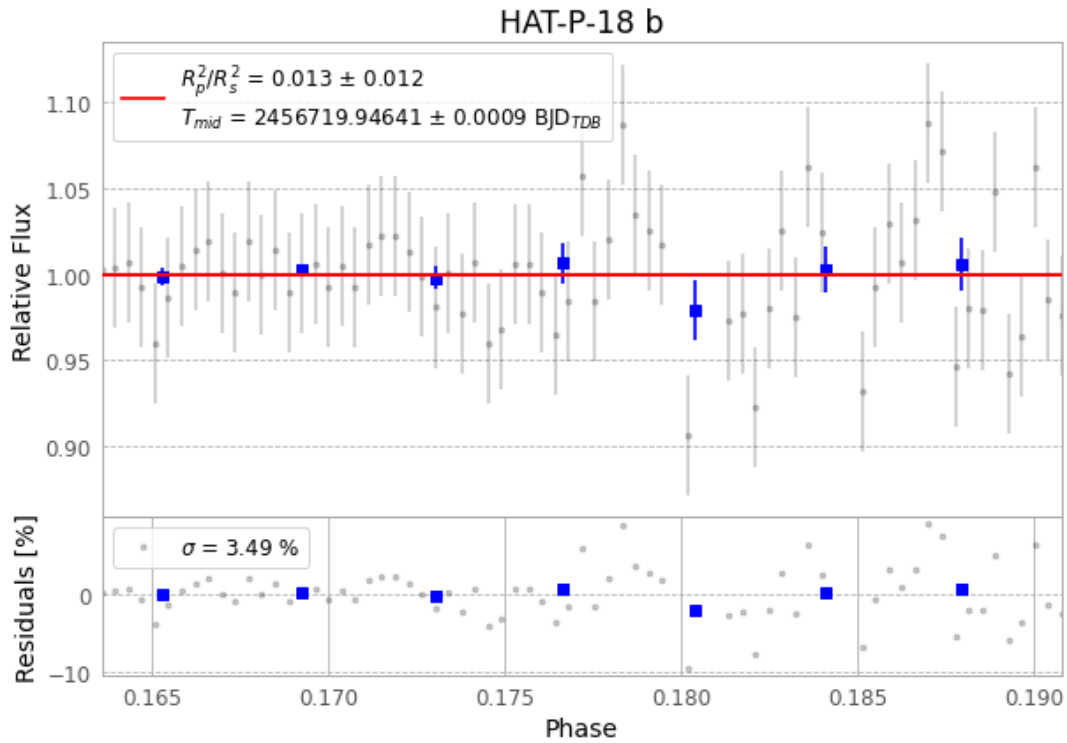


Figure 1 — FinalLightCurve_HAT-P-18 b_2014-03-04.png (SHA-256 3e5b9f352c82ff42...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [226, 202], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 379e43f4be486c7b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

73 FITS frames (48 MB), HATP-18140304081812.FITS → HATP-18140304115412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-18-140303.log (e4787fcfb6ae...), FinalLightCurve_HAT-P-18 b_2014-03-04.png (3e5b9f352c82...), FinalLightCurve_HAT-P-18 b_2014-03-04.csv (4fa778d309e1...)

Compute resources

Wall time 369.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-16 01:11Z.